

Assignment Specification

This assignment aims to develop a conversational system with deep knowledge about hotels. This knowledge comprises instance-level data (hotel descriptions as structured data) and an underlying ontological model. To answer a question or fulfill a request, the conversational system performs queries against the knowledge base (already provided data.ttl + your ontology created in pairs) to obtain accurate background information. Then, using a GraphRAG-like approach, it leverages this information to produce an answer to a user question.

Assignment input

1. Hotel knowledge graph (data.ttl)
2. An example set of dialogues (examples.txt)
3. A Quick Start Jupyter notebook (psai1-student-notebook.ipynb)

Assignment steps

1. **Analyze the provided knowledge graph (data.ttl)** For the KG, explore how many entities there are, how they are described, and what the different properties mean. To assist you in querying the data, you can use the OWLAPY Python library, which provides useful functionalities for working with ontologies. For this part of the assignment you can use the provided Jupyter notebook (psai1-student-notebook.ipynb) that includes installation instructions and the general structure of the assignment.
2. **Create a small ontology that can support expressive inference about hotels and analyse the dialogues (examples.txt).** The ontology does not need to be too large, and should focus on enabling meaningful inferences. You can either make it from scratch or extend the provided ontology file (data.ttl) with TBox information. It is recommended to do it in [Protege](#). The goal is to ensure that your ontology can support logical reasoning and provide insights based on the data. The deliverable for this step is your completed ontology file, either as a new file or an updated version of the provided data.ttl file. Once you have created your ontology, use it as the foundation for designing dialogues. Study the examples in examples.txt to understand their structure and content. Then, create 10 dialogues of your own, ensuring a range of difficulty levels: 5 easy ones and 5 more challenging ones. These dialogues should illustrate how your ontology can support reasoning and should include references to the types of information modeled in your ontology. Consider the following while creating your dialogues:
 - What kind of information mentioned in your dialogues is supported by the ontology?
 - How can reasoning with the ontology help in understanding or answering questions about the dialogues?
3. **Deploy a reasoning environment** that combines the ontology with the KG data, treated as the ABox in the OWL knowledge base. The idea is that instance queries

with complex class expressions should be used to retrieve different hotels, where reasoning is crucial for many aspects. For example, a query "give me places that are close to a coast" would also return places next to a beach if the query is evaluated together with the ontology that states that a place next to the beach is next to a coast.

4. **Instruct a LLM (i.e.: Llama3.2) to produce the query or components of the query (e.g., keywords) against the KG** based on the conversation so far and the known graph structure. The input should be a set of user questions, which are the dialogues you designed above. The output should be in some format that can be used by your reasoner.
5. **Use the LLM to summarize some result into a natural language response to the user.** First of all, use the reasoner to infer knowledge from the generated queries (and make sure the queries are valid beforehand). Then use a KG-augmented RAG formalization for the final output, where the model gets the types and an instruction and generates text.
6. **Evaluate your approach** using a text generation metric. For the questions with an answer of sets, you can compute an intersection-over-union score. You can also manually evaluate the generated queries by comparing with the ones you wrote in the dialogues, or come up with your own evaluation metrics.
7. **Write a report** of 5 pages in [LNCS](#) format consisting of:
 - a. an introduction to the challenge that you are tackling
 - b. your analysis of the knowledge graph
 - c. a description of your method, with your reasoning through the previous steps
 - d. discussion of the results of your evaluation
 - e. a discussion of lessons learned about the strengths and weaknesses of your approach

Assignment results

- A small-scale integrated logic-enhanced conversational system that can answer natural questions by converting them into queries against a KG+ontology system (30% of the grade)
- A report that reflects on the process and the lessons learned (70 % of the grade)

Intended learning outcomes

1. Consolidate the knowledge acquired about conversational AI and formal logic in AI, by bridging the gap between conversational systems and logical reasoning.
2. Learn to leverage structured data and ontologies to enable grounded reasoning by LLMs in conversational settings.

Necessary resources

These are our recommendations, following discussion you could go a different route but that would also mean that we can offer less support.

- Environment with GPU access - Kaggle notebooks (recommended), Colab notebooks or local.
- Libraries for conversational AI in Python (ollama)
- To access the ontology and reasoner: Protege, OWLAPY